

GATE-DA

Subject – (Artificial Intelligence)

Mock Test – 1

Topics Covered

Search: Informed, Uninformed

Adversarial; Logic, Propositional, Predicate;

Reasoning under uncertainty topics — conditional independence representation

Exact inference through variable elimination

Approximate inference through sampling.

Section A – Artificial Intelligence

Q1 (1 Mark, MCQ)

For a search tree with branching factor b and shallowest goal depth d , the worst-case time complexity of BFS is:

A. $O(bd)$

B. $O(b^d)$

C. $O(d^b)$

D. $O(\log b)$

Q2 (2 Marks, MSQ)

Which algorithms are complete for finite branching factor?

A. BFS

B. DFS

C. Iterative Deepening Search

D. Uniform Cost Search (positive costs)

Q3 (2 Marks, MCQ)

A heuristic $h(n)$ satisfies:

$$h(n) \leq h^*(n)$$

for all nodes.

This heuristic is:

- A. Consistent only
- B. Admissible only
- C. Both admissible and optimal
- D. Neither

Q4 (2 Marks, NAT)

A complete search tree has branching factor 4 and depth 5.

Total leaf nodes = _____

Q5 (2 Marks, MSQ)

Which statements about A* are true?

- A. Admissible heuristic guarantees optimality.
- B. Consistent heuristic implies admissibility.
- C. A*=UCS when $h(n)=0$.
- D. A* always expands fewer nodes than UCS.

Q6 (2 Marks, MCQ)

The major drawback of BFS is:

- A. Incompleteness
 - B. Non-optimality
 - C. High memory consumption
 - D. Infinite loops
-

Q7 (2 Marks, NAT)

A search tree has branching factor 3 and depth 6.

Total nodes generated including root = _____

Q8 (2 Marks, MSQ)

Which search algorithms can be optimal?

- A. BFS (unit costs)
 - B. UCS
 - C. A* (admissible heuristic)
 - D. Greedy Best First Search
-

Q9 (2 Marks, MCQ)

A heuristic h_1 dominates h_2 if:

- A. $h_1(n) \leq h_2(n)$
 - B. $h_1(n) \geq h_2(n)$
 - C. $h_1(\text{goal}) > 0$
 - D. h_2 inadmissible
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Q10 (3 Marks, MCQ)

For A* graph search:

A heuristic is admissible but inconsistent.

Which statement is correct?

- A. Optimality is impossible.
 - B. Optimality holds only with node reopening.
 - C. A^* =Greedy Search.
 - D. Consistency and admissibility are equivalent.
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Q11 (2 Marks, MCQ)

Minimax assumes:

- A. Cooperative opponent
 - B. Rational opponent
 - C. Random opponent
 - D. Unknown opponent
-

Q12 (2 Marks, NAT)

A complete binary game tree has depth 8.

Number of terminal nodes = _____

Q13 (2 Marks, MSQ)

Alpha-Beta pruning:

- A. Changes minimax value
 - B. Reduces node expansion
 - C. Depends on move ordering
 - D. Can prune entire subtrees
-

Q14 (2 Marks, MCQ)

Best-case Alpha-Beta complexity:

- A. $O(b^m)$
 - B. $O(b^{m/2})$
 - C. $O(mb)$
 - D. $O(m^2)$
-

Q15 (3 Marks, MCQ)

For branching factor b and depth m ,

Minimax complexity is:

- A. $O(bm)$
 - B. $O(m^b)$
 - C. $O(b^m)$
 - D. $O(\log b)$
-

Q16 (2 Marks, MCQ)

$$P \rightarrow Q$$

is equivalent to:

- A. $P \vee Q$
 - B. $\neg P \vee Q$
 - C. $\neg Q \vee P$
 - D. $P \wedge Q$
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Q17 (2 Marks, MSQ)

Identify tautologies.

A. $P \vee \neg P$

B. $P \wedge \neg P$

C. $(P \rightarrow Q) \vee (Q \rightarrow P)$

D. $P \leftrightarrow P$

Q18 (2 Marks, NAT)

For 8 propositional variables,

number of rows in truth table = _____

Q19 (2 Marks, MCQ)

Resolution requires formulas in:

A. DNF

B. CNF

C. NNF

D. Horn form only

Q20 (3 Marks, MCQ)

Resolution derives contradiction by proving:

A. Goal

B. Negated goal

C. Empty clause

D. CNF

Q21 (2 Marks, MSQ)

Components of FOL include:

- A. Constants
 - B. Variables
 - C. Predicates
 - D. Quantifiers
-

Q22 (2 Marks, MCQ)

Which quantifier denotes existence?

- A. $\forall$
 - B. $\exists$
 - C. $\Rightarrow$
 - D. $\wedge$
-

Q23 (2 Marks, NAT)

A KB contains 6 independent Boolean predicates.

Possible interpretations = _____

Q24 (3 Marks, MCQ)

The process of finding substitutions that make two predicates identical is:

- A. Resolution
 - B. Skolemization
 - C. Unification
 - D. Grounding
-

Q25 (2 Marks, MCQ)

In Bayesian Networks:

- A. Cycles allowed
 - B. DAG required
 - C. Undirected graph
 - D. Complete graph
-

Q26 (2 Marks, MSQ)

Markov Blanket of node X consists of:

- A. Parents
 - B. Children
 - C. Children's parents
 - D. Grandparents only
-

Q27 (2 Marks, NAT)

A BN variable has 3 binary parents.

CPT entries required = _____

Q28 (2 Marks, MCQ)

Variable Elimination improves inference by:

- A. Enumerating more states
 - B. Reusing intermediate factors
 - C. Ignoring evidence
 - D. Sampling
-

Q29 (2 Marks, MSQ)

Approximate inference methods include:

- A. Rejection Sampling
 - B. Likelihood Weighting
 - C. Gibbs Sampling
 - D. Resolution
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Q30 (3 Marks, MCQ)

Suppose A and B are conditionally independent given C.

Which is always true?

A.

$$P(A, B \mid C) = P(A \mid C)P(B \mid C)$$

B.

$$P(A, B) = P(A)P(B)$$

C.

$$P(A \mid B) = P(A)$$

D.

$$P(C \mid A) = P(C \mid B)$$

Section B – Aptitude

Q31 (1 Mark, MCQ)

Choose the correct word:

The argument was logically _____.

A. coherent

B. coherence

C. coherently

D. cohering

Q32 (1 Mark, NAT)

If $x + y = 14$ and $xy = 45$,

find $x^2 + y^2$.

Q33 (1 Mark, MCQ)

Find the next term:

1, 4, 9, 16, 25, ?

A. 30

B. 35

C. 36

D. 49

Q34 (2 Marks, MCQ)

A train of length 180 m crosses a platform of length 220 m in 20 seconds.

The speed of the train is:

A. 18 m/s

B. 20 m/s

C. 22 m/s

D. 24 m/s

Q35 (2 Marks, MCQ)

A product is marked 25% above cost price. A discount of 10% is given.

The profit percentage is:

- A. 10%
 - B. 11.5%
 - C. 12.5%
 - D. 15%
-

Q36 (2 Marks, MSQ)

Which of the following numbers are prime?

- A. 41
 - B. 43
 - C. 45
 - D. 47
-

Q37 (2 Marks, NAT)

A can complete a work in 12 days and B can complete the same work in 18 days.

Working together, they complete the work in _____ days.

(Round to 2 decimal places)

Q38 (2 Marks, MCQ)

The average of 8 numbers is 15.

If one number 27 is removed, the average of the remaining numbers becomes:

- A. 12.86
- B. 13.29

C. 14.14

D. 14.71

Q39 (2 Marks, MCQ)

A box contains 5 red, 4 blue and 3 green balls.

Two balls are drawn randomly without replacement.

Probability that both are red:

A. $\frac{5}{33}$

B. $\frac{10}{66}$

C. $\frac{20}{132}$

D. $\frac{15}{66}$

Q40 (2 Marks, MCQ)

Five friends P, Q, R, S and T sit in a row.

- P sits immediately to the left of Q.
- R sits somewhere to the right of Q.
- S sits at one of the ends.
- T is not adjacent to R.

How many valid arrangements are possible?

A. 8

B. 10

C. 12

D. 16